



Society of Petroleum Engineers

SPE-227700-MS

Application of Machine Learning for Accelerating the Fault Interpretation in the Structurally Complex Area: A Case Study of the Sirte Basin, Libya

A. Alzaruk, SLB, Tripoli, Libya; A. Gliwan, A. Alqadhi, and A. Ajaj, Petroleum Research Center, Tripoli, Libya; A. Elakuri, SLB, Tripoli, Libya

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227700-MS](https://doi.org/10.2118/227700-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

The Sirte rift basin in Libya is a complex geological structure characterized by a series of tilted normal fault blocks formed during the early late Cretaceous. This study presents a novel application of machine learning (ML) assisted fault interpretation in the Sirte Basin, focusing on the western part of the rift basin where numerous significant oil fields are located. The integration of ML algorithms with seismic interpretation enables rapid and accurate identification of complex fault structures, reducing evaluation time and improving interpretation accuracy. This study demonstrates the potential of ML assisted fault interpretation to improve subsurface understanding, particularly in structurally complex areas. The proposed workflow leverages pre-trained ML models to predict faults from 3D seismic volumes, which are then compared to user-interpreted fault identification supported by seismic variance attributes. The results show reasonably acceptable accuracy, with an improved fault prospect volume that overtakes variance technique. This study contributes to the development of innovative seismic interpretation workflows, enabling geoscientists to better understand subsurface structures and make more informed decisions. The proposed ML assisted fault interpretation workflow has the potential to be applied to other geological settings, reducing time and improving accuracy in fault prediction.

Challenges in seismic fault interpretation, 3D seismic imaging often fails to provide sharp imaging of faults, leading to ambiguity regarding fault geometry and continuity. To address this challenge, we employed a combination of manual interpretation and deep learning-based automatic fault detection using a pre-trained 3D convolutional neural network (3D CNN) algorithm. The 3D CNN model displayed remarkable similarities to the manually picked faults.

Introduction

The Sirte Basin is a northwest oriented wide Meso-Cenozoic rift basin located in northcentral Libya which covers an area of approximately 600,000 km² that is bordered to the east by the block-faulted region of the Cyrenaica platform, including Al Jabal al Akhdar. It is bounded to the south by the Tibesti Mountains the Al Kufra basin to the southeast, to the southwest by the Nafusah Uplift and Al Qargaf arch and the Ghadames and Marzuq Basins, with the Mediterranean Sea to the north ([Fig.1](#)).

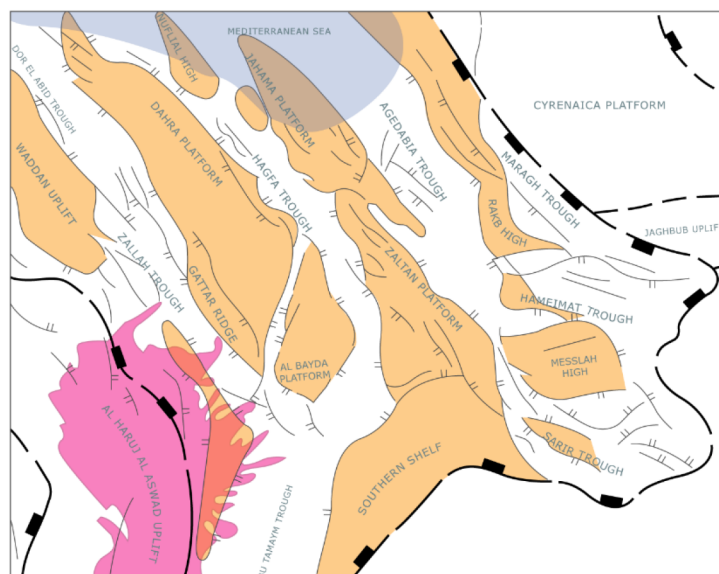


Figure 1—Tectonic elements of the Sirte rift basin.

The Sirte rift basin consists of a series of tilted normal fault blocks which resulted from northeast southwest extension tectonics during the early late Cretaceous. Special emphasis devoted to studying the western part of the rift basin due to well manifesting rift architecture and large numbers of significant oil fields.

The dataset used in this study is a standard prestack time migration (PSTM) amplitude volume, which shows how underground structures reflect seismic waves. It was used in its original form without extra modifications. This dataset helps detect faults by analyzing changes in amplitude. To manually identify faults, seismic variance attributes were used, which highlight areas with high differences in amplitude. The process involves carefully examining different sections of the seismic volume to find fault lines. A total of 64 faults were identified. These manually detected faults were then used as a reference to test how well machine learning methods can detect faults (Fig.2).

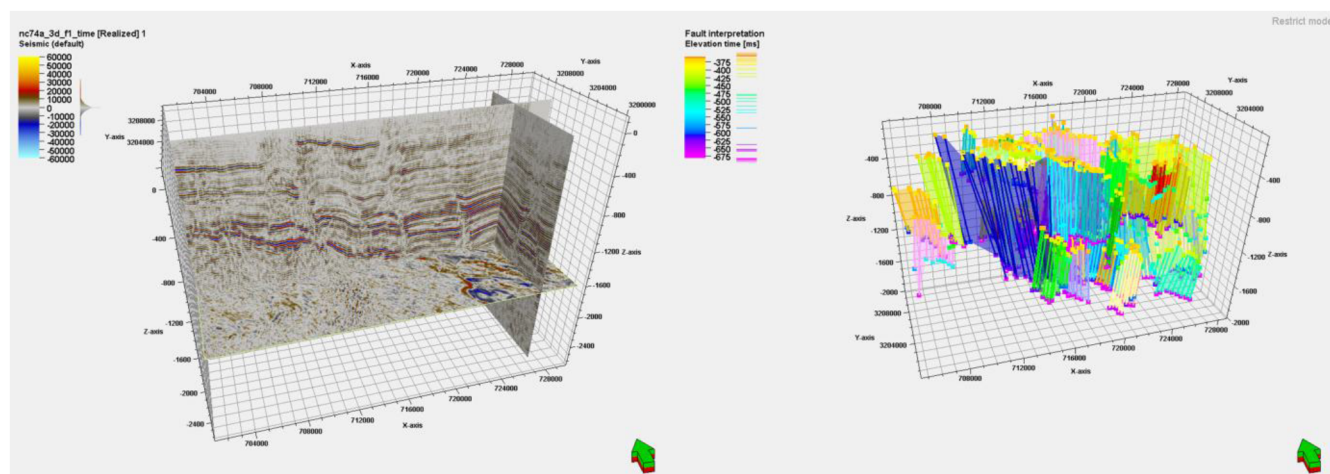


Figure 2—The seismic cube that covers the study area with results of manual fault interpretations.

Methodology

Prediction capabilities are performed by what is referred to as a prediction 'brain'. This is a result of learning through data, using the 3D convolutional neural networks (3D CNN), where the system is trained to identify

faulted discontinuities using a series of expertly labeled seismic images with faults positively identified, also known as training labels. When trained, a prediction model can then be used to perform predictions elsewhere, at unseen locations, within a given dataset.

These techniques enable the prediction of faults on a voxel-by-voxel basis, output as a fault probability attribute cube, revolutionizing the concept of a seismic based fault cube. This enables entirely new levels of accuracy and efficiency that provides a robust basis for fault interpretation, extraction, modeling, and validation (Fig.3).

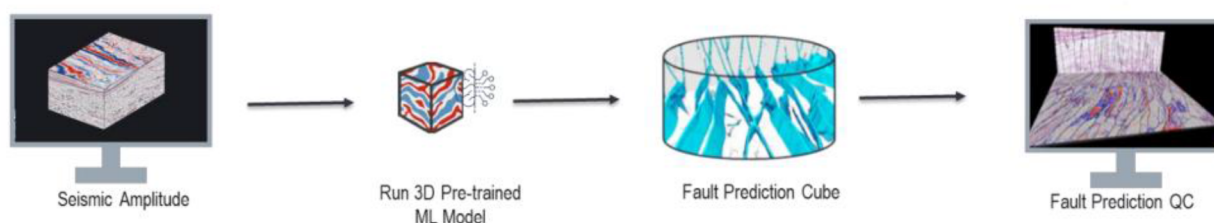


Figure 3—Machine learning 3D pre-trained fault prediction workflow.

The 3D pre-trained fault interpretation is a supervised predication model. It has been trained using 3D CNN network using synthetic data. The example of synthetic data is presented in (Sarajaervi, 2020) paper.

Using the training network, interference predication is run at different seismic resolution before margining the result (Goledowski, 2022).

The present work shows the use of machine learning to reduce the interpretation time of geological faults from seismic data, an activity that consumes a considerable amount of the interpreter's time, which is not characterized by a simple mechanical activity. It requires prior knowledge of the structural style associated with the interpreted area, requiring expertise from the interpreter to properly deal with the limitations that the seismic data normally imposes in terms of resolution and quality. The 3D CNN architecture applied here is a variation of U-Net, developed in 2015 with an initial application focused on segmentation in medical images (Ronneberger, 2015) and image processing and pattern recognition (Badrinarayanan, 2017). The recent success of ML approaches has brought new insights into the domain of seismic interpretation, capturing more structural features, in an optimum time frame. This workflow is based on the synergy of the interpreter's expertise uses the advanced image recognition technology (3D CNN) to estimate a fault prediction volume. This volume is then used in the extraction of faults objects directly from seismic data.

A supervised deep learning method was used with a 3D Convolutional Neural Network (CNN) that was pre-trained on various synthetic seismic datasets. The model was designed to detect faults by recognizing patterns in the seismic data and identifying sudden changes in amplitude that indicate fault locations. The CNN has multiple layers that gradually extract important details about the fault structures. The fault detection process involved several steps to improve accuracy. First, the 3D CNN analyzed the seismic data to create a "fault probability cube," where each point was given a score showing how likely it was to be part of a fault (Fig.4).

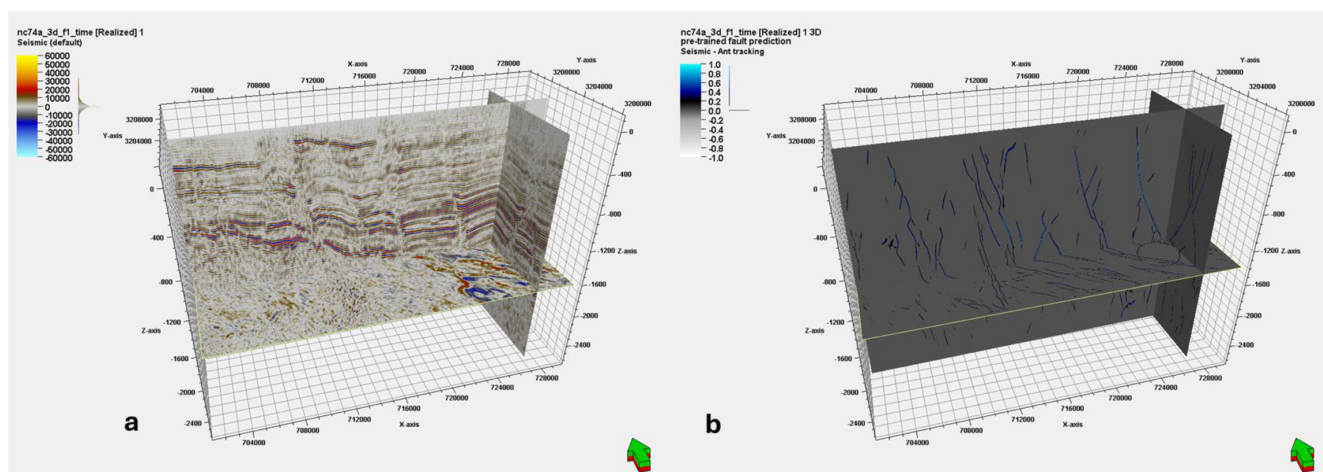


Figure 4—(a) The original seismic cube. (b) Machine learning 3D pre-trained fault prediction cube.

To evaluate the results, two attributes were created a planarity cube and azimuthal cube. Planarity attributes helped ensure the detected faults were continuous and structurally correct while reducing errors caused by noise in data. A planarity cube is used to extract and split faults at the intersections. It highlights how planar a fault region is with a value range from 0 (no flatness found in the search radius) to 1 (a completely flat plane in the search radius). (Fig.5).

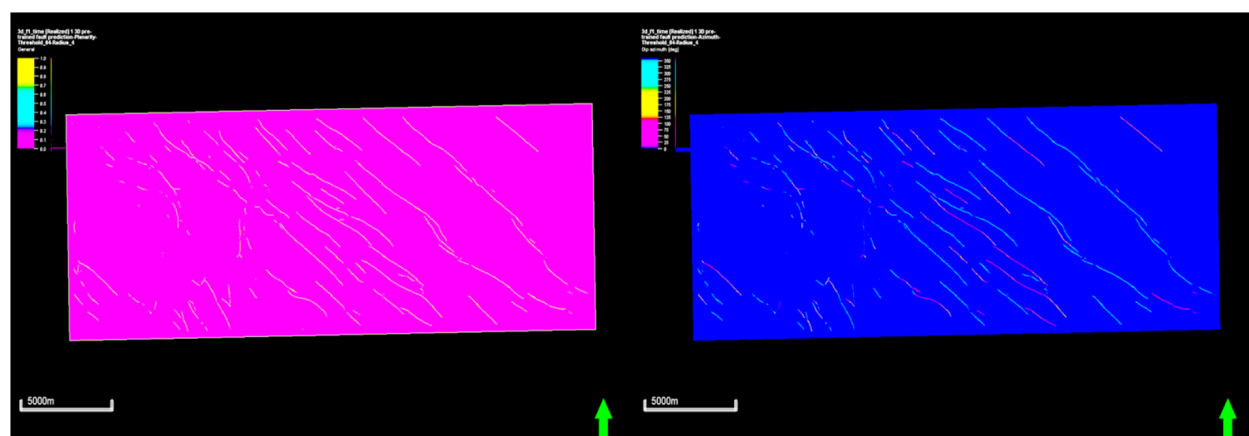


Figure 5—Planarity and azimuth attributes generated for the faults

Azimuthal attributes were used to highlight differences in direction, making it easier to distinguish real faults from other geological features. The azimuth cube is used to guide the extraction of the faults. In conjunction with the azimuth sectoring, its facilitates the extraction of the faults with a consistent azimuth. This means that the azimuth of an extracted fault only changes slowly and therefore, provides for a geologically consistent fault. The azimuth is calculated according to the geological definition of the strike and defined clockwise from north in 360 degrees.

To validate and interpret the results which is fault pointsets, a mixer cube was created by blending the original seismic volume with the predicted fault probability cube, enabling geophysicists to visually analyze fault locations in relation to raw seismic data (Fig.6). This step allows for qualitative verification of the machine learning models output by overlaying fault predictions onto the seismic reflections.

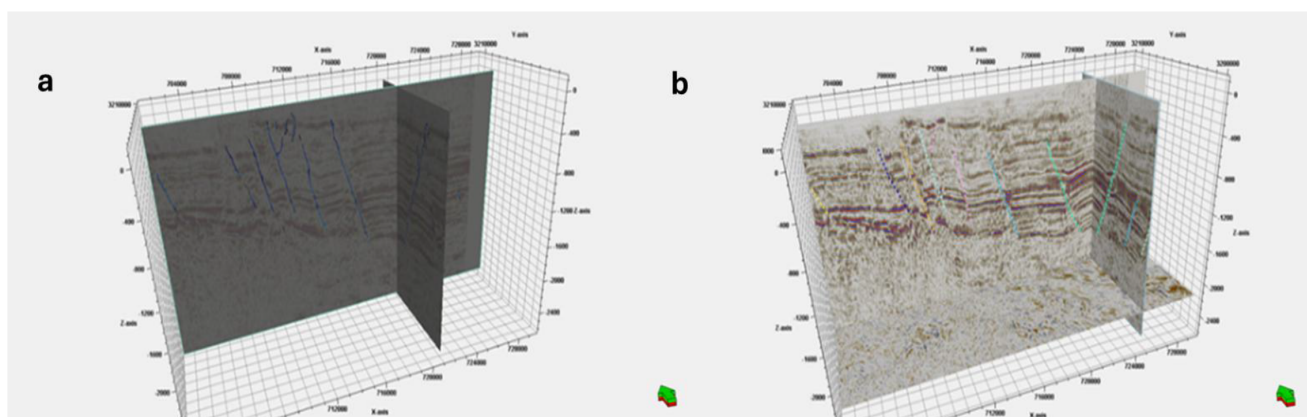


Figure 6—(a) Mixer of original seismic with fault cube. (b) original seismic cubes with ML fault interpretations results.

Additional post-processing steps were applied to refine the fault probability cube, including thresholding to filter out low-confidence fault predictions, morphological operations to enhance the continuity of fault surfaces, and connectivity analysis to remove isolated artifacts that do not represent geologically meaningful structures. From the refined fault volume, discrete fault points were extracted and clustered to reconstruct fault planes, providing a structured dataset for further structural interpretation.

Results and Discussions

Efficiency and Scalability: Accelerating the Fault Interpretation Timeframe

The automated fault identification using the pre-trained CNN significantly reduced interpretation time compared to manual fault identification. This approach would be the preferred method in future interpretation projects, especially for large-scale seismic datasets. The significant time savings achieved by using ML-assisted fault interpretation methods. The ML method took less than 20% of the evaluation time required by manual methods to complete. Moreover, the ML assisted fault interpretation outputs showed overall better quality than manual faults. However, the ML assisted method did not capture as many faults as the manual method, particularly in areas where the seismic expression of faulting was poor.

The integration of machine learning into fault interpretation workflows has revolutionized the way interpreters analyze and understand complex subsurface structures. A key benefit of this technology is its potential to significantly reduce evaluation time, enabling interpreters to focus on high-level fault features and complex data areas. A comparative study of manual and ML assisted fault interpretation workflows revealed substantial time savings. The ML workflow reduced evaluation time for major fault features. This considerable reduction in interpretation time enables experts to allocate more resources to challenging areas of the data, where ML a method may face limitations due to data quality or other factors. The increased efficiency of the ML workflow is attributed to its ability to quickly and accurately identify major fault features. With ML assistance, interpreters can focus on refining their understanding of complex fault networks, rather than spending excessive time identifying primary fault features. Moreover, the ML workflow enables interpreters to explore more scenarios, test multiple hypotheses, and refine their understanding of the subsurface. Within the scope of faults that can be derived from ML workflow, it was observed that results were not just quicker but better quality overall. The improved accuracy and precision of the ML workflow is attributed to its ability to analyze large datasets, identify patterns, and learn from experience.

Assessing the Value of ML-Assisted Fault Interpretation by comparison with Traditional Approach

To evaluate the effectiveness of the ML approach, its output was compared with manually interpreted faults. This comparison aimed to quantify the value contribution of deploying ML assisted fault interpretation.

The assessment focused on key aspects as the degree of overlap between ML derived faults and manual fault picks. The reference fault dataset used for validation consisted of previous 64 manually interpreted faults, carefully selected to correspond to noticeable fault features on seismic amplitude volumes used for the project.

A prediction match analysis was conducted to compare the overlap between ML derived faults and previous manually interpreted faults (Fig.7). A match was deemed valid where there was a perfect overlap between the two interpretations. The results showed that most of 64 manually picked faults were associated with visible fault features on seismic, these manual faults good matched with ML derived faults, indicating that the ML system predicted of faults that were manually picked prior to commencement of the ML project. This demonstrates the effectiveness of the applied ML workflow for fault interpretation. Furthermore, the ML System also yielded valid faults that were not previously picked manually, highlighting its potential to identify new fault features (Fig.8).

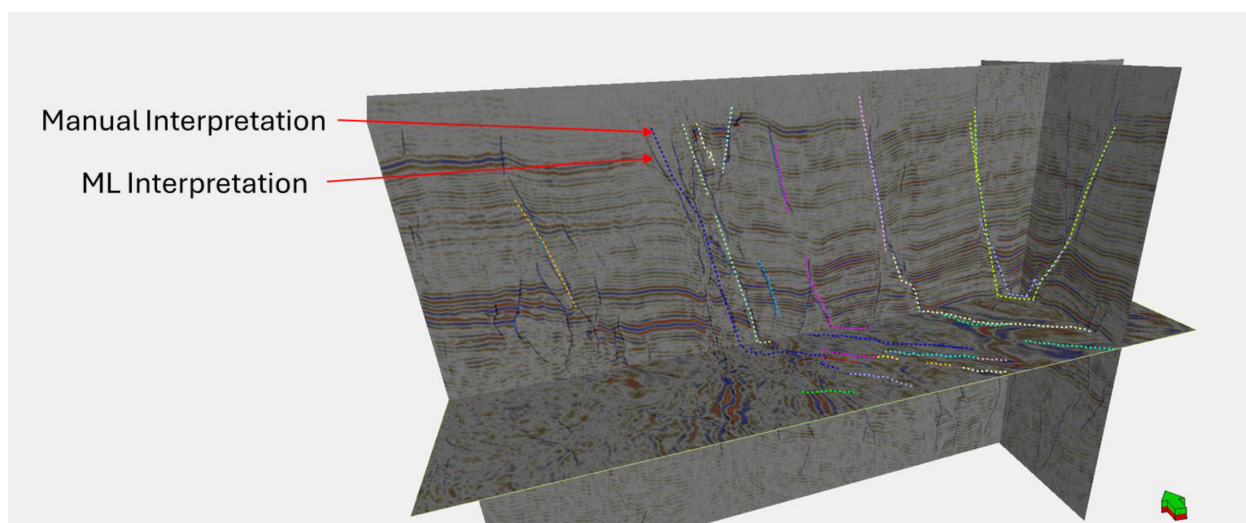


Figure 7—shows an example of a matching ML approach derived fault and manually interpreted fault. The match between the two faults is particularly good, demonstrating the accuracy of the ML-assisted method.

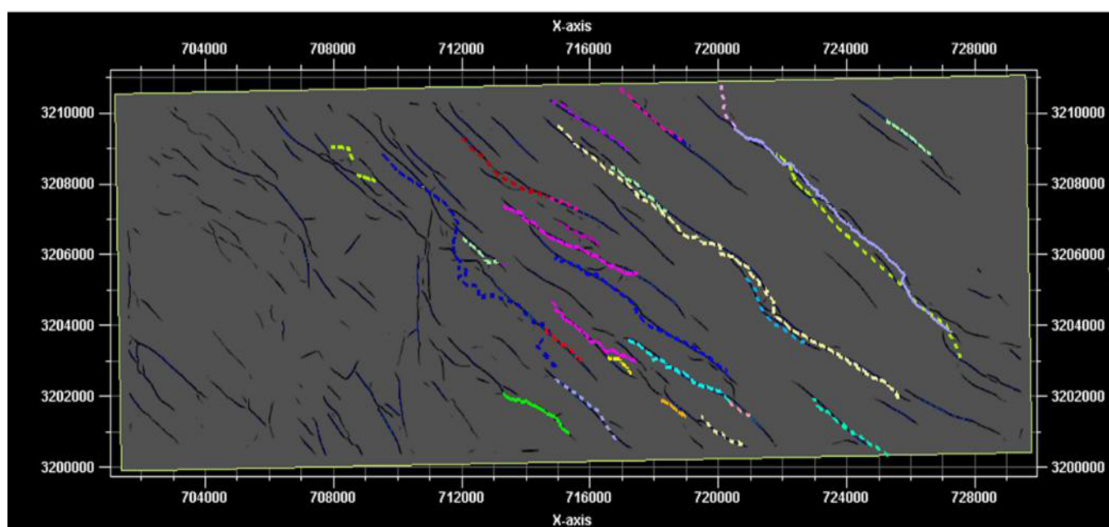


Figure 8—Time slice shows results of ML assisted fault interpretation (fault prediction cube) overlaid by manually interpreted fault (color dash lines). The ML method provides more accurate and detailed fault predictions, demonstrating its superiority over traditional analysis.

ML comparison with traditional fault detection by variance attribute

As the traditional fault detection techniques guided by seismic variance attributes can be time-consuming and require high interpretation skills and structural geology domain experience. This method may also struggle to identify subtle or complex faults, leading to incomplete or inaccurate interpretations. A comparative with ML predicted faults revealed that the 3D CNN based approach provided cleaner fault anomalies and captured subtle faults that were difficult to map using traditional methods.

As expected, clear continuous northeast southwest trending faults are seen on the ML fault cube, whereas it appears as disconnected segments and not extending far enough on the variance cube. The vertical sections demonstrate the clarity of event terminations for fault detection on ML cube but unclear on the variance cube. There are some highlights cases the value of ML identified lineaments that are not detected by the variance (Fig.9).

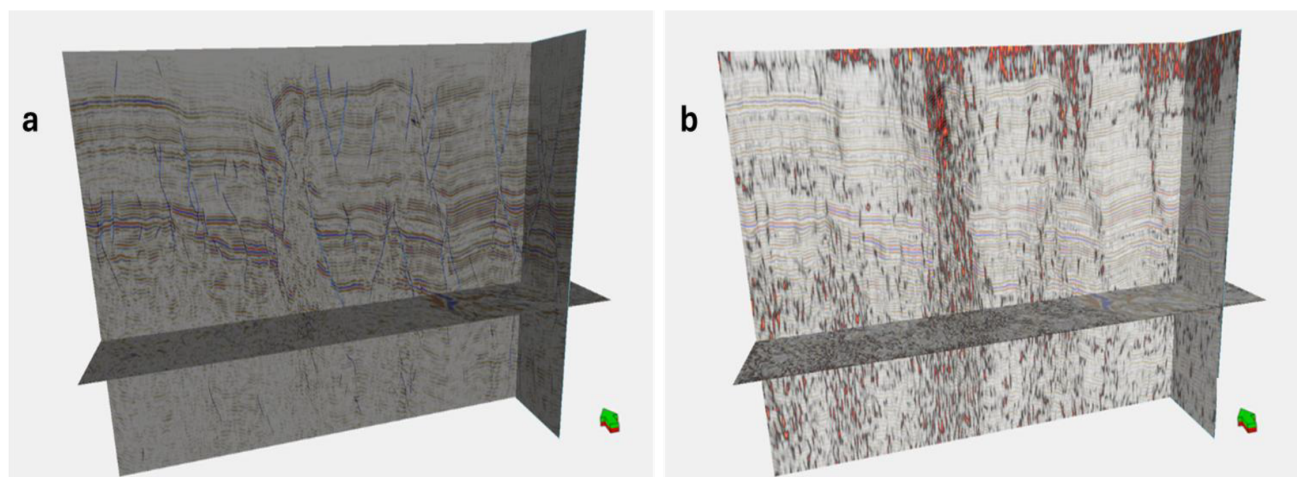


Figure 9—comparison between (a) ML fault prediction cube with input seismic cube and (b) variance attribute cube with input seismic cube.

Faults identified by the ML pre-trained model were expressed better than those guided by seismic variance attributes, although most major faults identified on ML are seen uncertain on the variance (Fig.10).

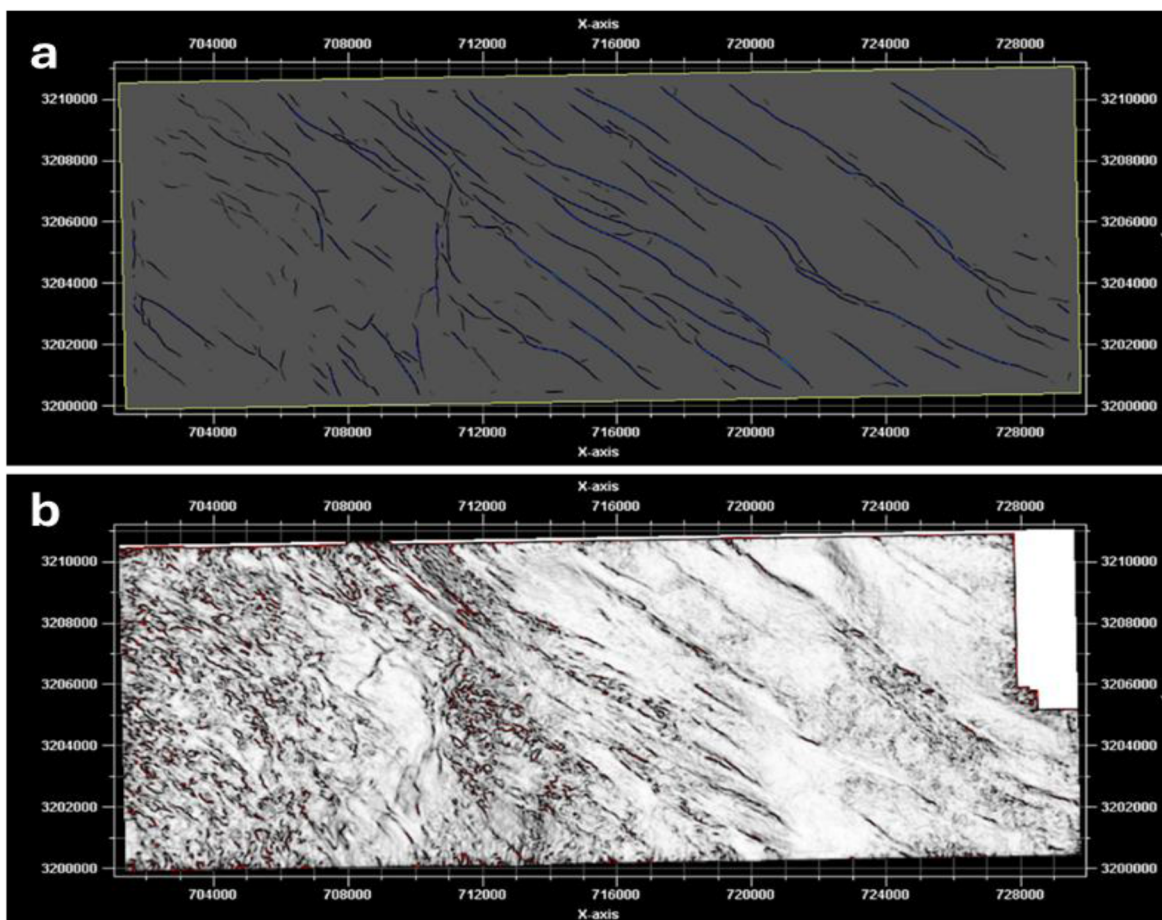


Figure 10—Comparison between (a) ML fault prediction and (b) variance attribute cube. The ML method provides more accurate and detailed fault predictions, demonstrating its superiority over traditional variance attribute analysis.

The Importance of Interpreter Judgement

The interplay between ML assisted fault interpretation and interpreter judgement is crucial in ensuring the accuracy and completeness of fault interpretation. An area where the ML approach did not make predictions due to the absence of evident fault indications on seismic data. Conversely, the interpreter manually picked faults across the same area, highlighting the importance of interpreter judgement in complementing ML assisted fault interpretation (Fig. 11). The highlights an area where manual faults were picked, but the ML did not detect any fault, indicating potential limitations of the ML method in areas with poor seismic data quality. In this case, the manually interpreted faults trusted more on finer interpreter judgement than data signature. This highlights the confident value of interpreter judgement in additional the results of ML assisted fault interpretation. The validation and comparison exercise demonstrated the effectiveness of the ML approach in predicting faults and improving workflow efficiency. The results highlight the potential of ML assisted fault interpretation to transform the way interpreters analyze and understand complex subsurface structures. This disparity underscores the critical role of interpreter judgement in complementing the results of ML-assisted fault interpretation. In this case, the interpreter likely relied on judgement from adjacent intervals, other data sources, or their expertise to identify faults that were not apparent on seismic data. This highlights the importance of human oversight and expertise in validating and refining the output of ML-assisted fault interpretation workflows. Moreover, this example demonstrates the potential for ML-assisted fault interpretation to augment, rather than replace, interpreter judgement. By providing a robust and accurate foundation for fault interpretation, ML-assisted workflows can free up interpreters to focus on higher-level tasks, such as validating and refining results, and making informed decisions based on their expertise and

judgement. The interplay between ML-assisted fault interpretation and interpreter judgement is crucial in ensuring the accuracy and completeness of fault interpretation.

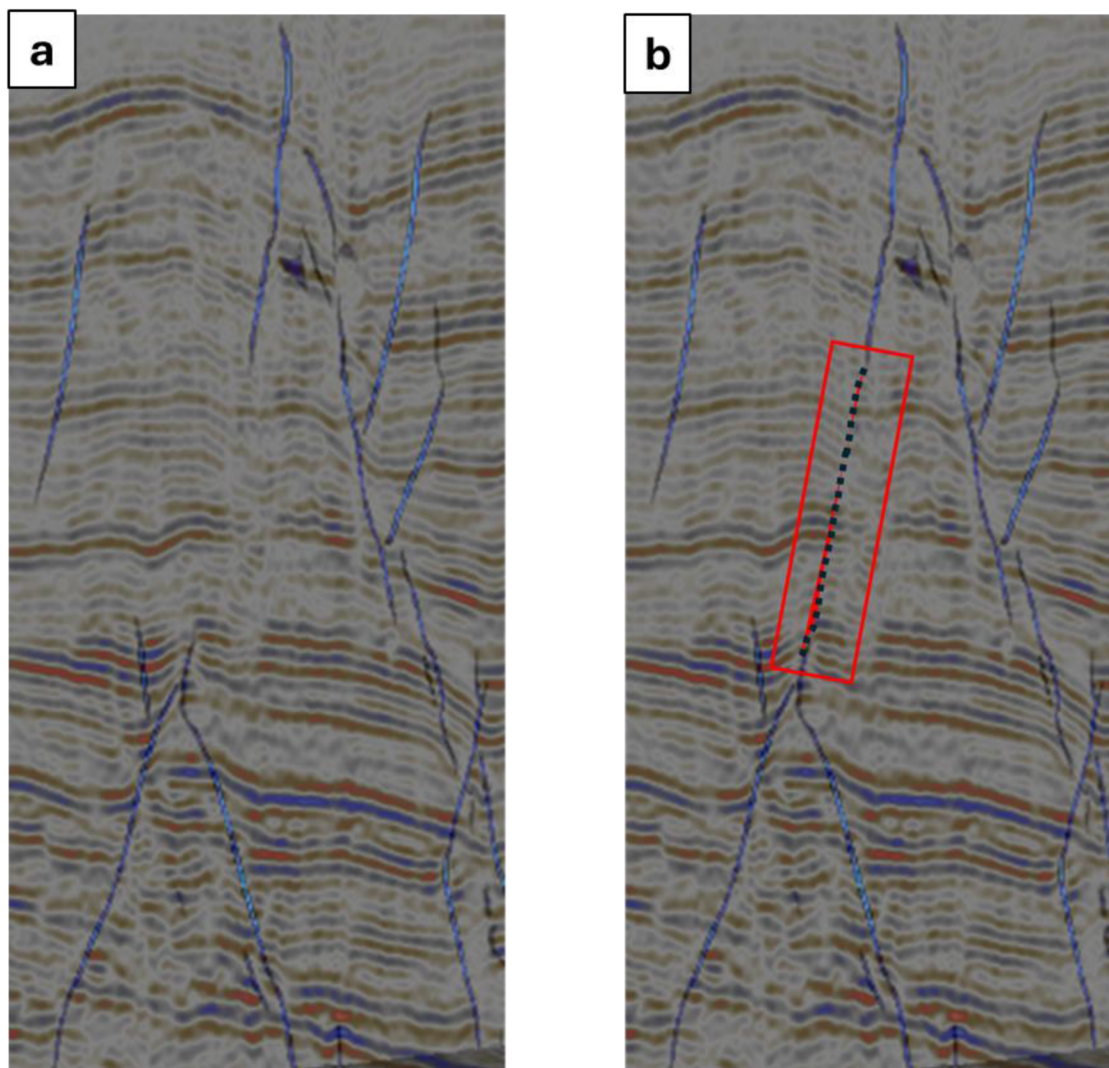


Figure 11—Illustrates an area where (a) the ML approach did not make predictions due to the absence of evident fault indications on seismic data. Conversely, (b) the interpreter manually picked faults across the same area, highlighting the importance of interpreter judgement in complementing ML assisted fault interpretation.

Conclusions

Accurate fault identification is essential for subsurface interpretation and reservoir characterization. The 3D CNN framework offers a promising approach for fault identification, but its effectiveness depends on the quality of the seismic dataset. Our results demonstrate that a pre-trained 3D CNN model can identify faults more accurately and efficiently than manual interpretation. The automated fault identification approach can significantly reduce interpretation time and is a preferred approach for future interpretation projects. This project successfully utilized machine learning (ML) to enhance fault interpretation in the Sirte basin, characterized by numerous near- vertical faults with small throws. The primary objectives were to increase efficiency and precision, both of which were achieved. By leveraging ML assisted fault interpretation, the project reduced the time needed to complete 65% of major fault interpretation to less than 20% of the time required by traditional methods. The integration of ML assisted fault interpretation with traditional methods yielded significant benefits. The ML method was used alongside traditional methods, which enhanced the benefits. The results of this project show that combining ML with traditional methods can be very effective.

By using ML to automate repetitive tasks, geoscientists can focus on more complex analysis and make better decisions. The successful application of ML assisted fault interpretation in this project demonstrates its potential to revolutionize subsurface characterization. As the oil and gas industry continues to evolve, the adoption of ML assisted fault interpretation can help geoscientists stay ahead of the curve. By embracing this technology, geoscientists can unlock new insights, improve decision-making, and ultimately drive business success. Ultimately, the benefits of ML assisted fault interpretation extend beyond the realm of subsurface characterization. By automating repetitive tasks and freeing up geoscientists to focus on higher-level analysis, ML assisted fault interpretation can help drive innovation and efficiency across the entire lifecycle. As industry continues to navigate the challenges of a rapidly changing energy landscape, the adoption of ML assisted fault interpretation can play a critical role in shaping the future of exploration and production.

Acknowledgments

Firstly, the authors want to express their greatest thanks to Libyan National Oil Corporation (NOC) and Petroleum Research Center (PRC) for their permission to use the data for this study. We also want to thank SLB for encouraging and supporting this project. Finally, the authors really appreciated the contributions of geosciences studies office staff in PRC for the very useful discussions and suggestions during the preparation of this technical paper.

References

- B. Goledowski, T.H. Bø, M. Sarajaervi, B.H. Fotland, H.G. Borgos and M. Nickel (2022) Overcoming the limited Field of View of convolutional neural network for fault detection: a multiscale approach. Second EAGE Digitalization Conference and Exhibition, Mar 2022, Volume **2022**, p.1–5.
- Cardoso da Silva, A. A., Paes de Almeida Junior, M., Pereira de Lima, K. T., Pesce, P. B. C., Martinez, C., Oliveira, G. M. B., Ferreira, R. N., & Caparica Junior, J. F. (2021, November 8-11). Assisted Interpretation of Geological Faults using Machine Learning and Cloud Computing: A Case Study in Pre-Salt environment. Proceedings of the 17 th International Congress of the Brazilian Geophysical Society held in Rio de Janeiro, Brazil, 8-11 November 2021. Petrobras & Schlumberger.
- Islam, M. M., Babikir, I., Elsaadany, M., Elkurdy, S., Siddiqui, N. A., & Akinyemi, O. D. (2023, October 14). Application of a Pre-Trained CNN Model for Fault Interpretation in the Structurally Complex Browse Basin, Australia. Proceedings of the Centre for Subsurface Imaging, Department of Geosciences, Universiti Teknologi PETRONAS.
- Oke, M., Tilley, S., Manral, S., Gacem, M. T., Mawlod, A., Al Daghar, K., Ben Jeddou, H., Mustapha, H., & Li, C. (2020, November 9). Application of Machine Learning-Assisted Fault Interpretation on Large Carbonate Field with Subtle Throws. Proceedings of the Petroleum Exhibition & Conference to be held in Abu Dhabi, UAE, 9-12 November 2020. ADNOC & Schlumberger.
- Ronneberger, O., Fischer, P. and Brox, T. (2015) U-net: Convolutional networks for biomedical image segmentation. In: International Conference on Medical image computing and computer-assisted intervention. Springer, 234–241.
- Badrinarayanan, V., Kendall, A. and Cipolla, R. (2017) SegNet: A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, **39**, 2481–2495.
- Sarajaervi, M., Hellem Bø, T., Goledowski, B., and Nickel, M. (2020) Robust Evaluation of Fault Prediction Results: Machine Learning Using Synthetic Seismic. Conference Proceedings, 1st EAGE Digitalization Conference and Exhibition, Nov 2020, Volume **2020**, 1–5 (Extended Abstract).